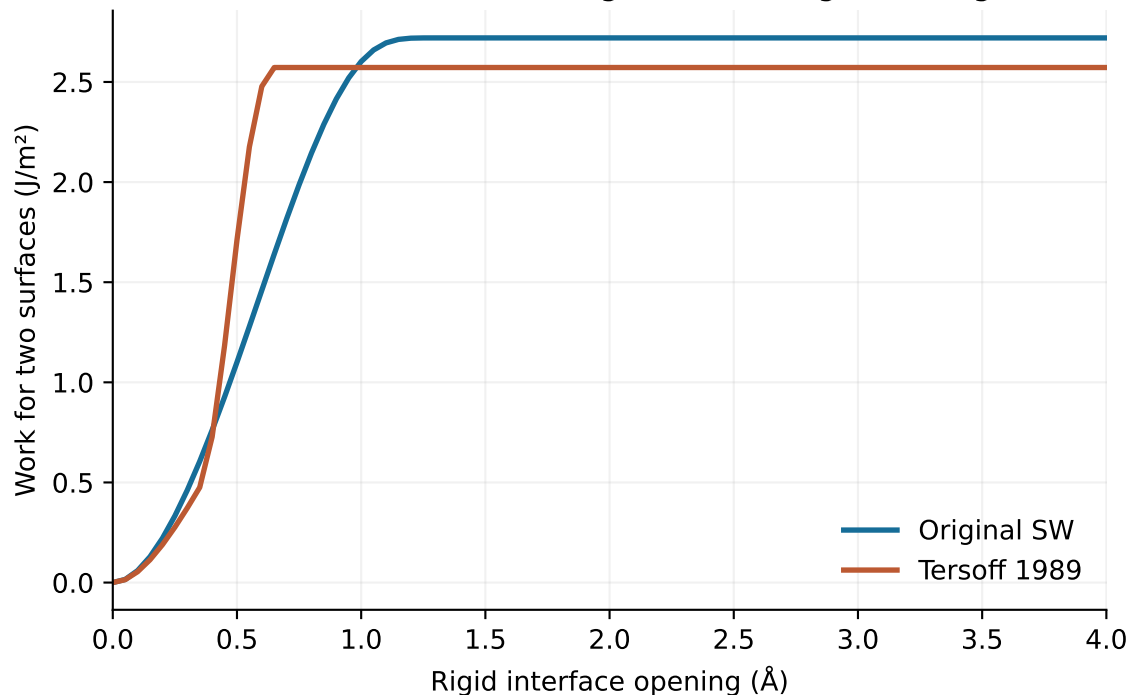
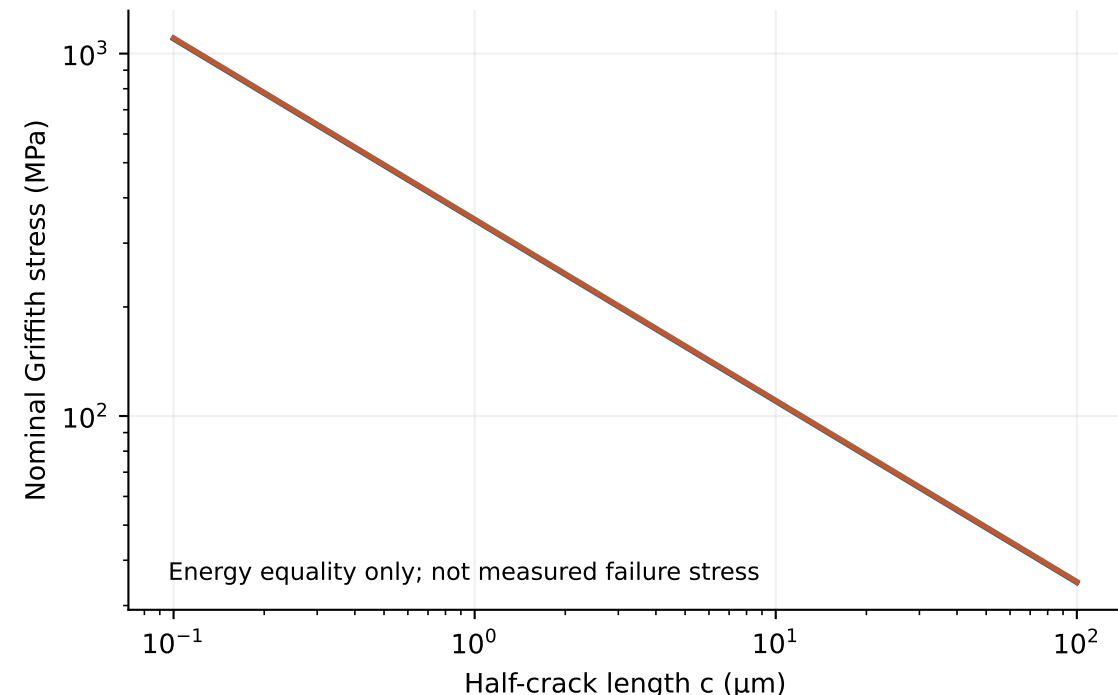


Si specimen loading bridge — static references, not material calibration

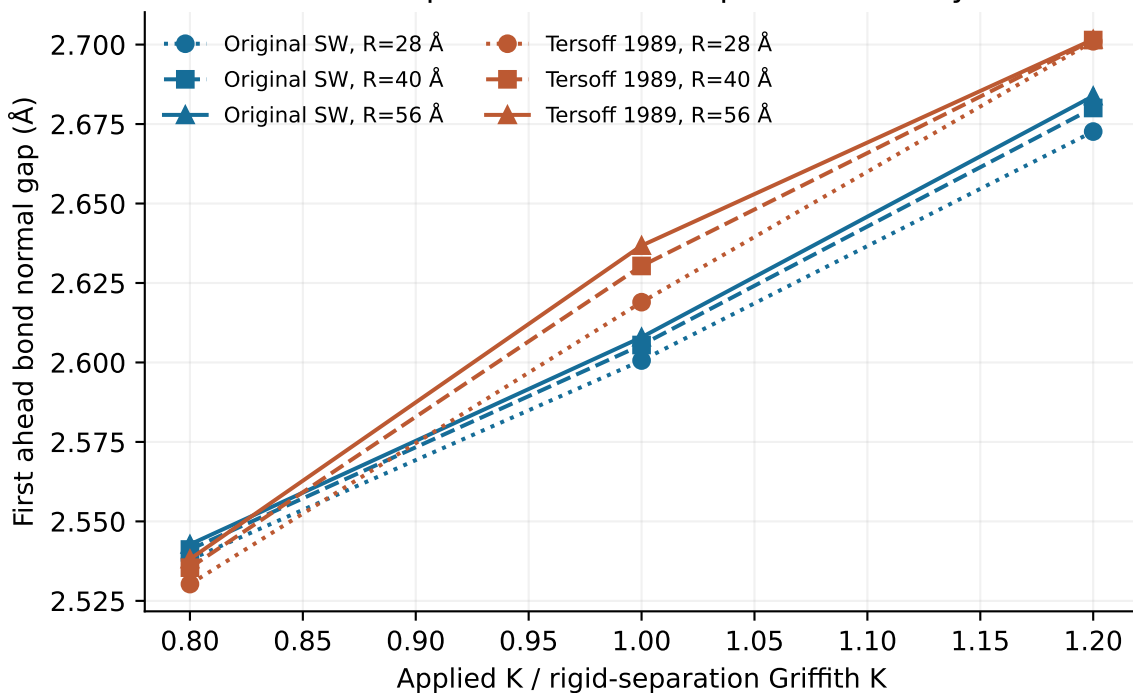
A. Same atomistic energies, no strength scaling



B. Infinite central crack: $Y=1$



C. Independent atoms / optical boundary



D. All free Cartesian directions tested

